

National Neighborhood Data Archive (NaNDA): Social Services by Census Tract, United States, 2003-2017



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Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains measures of the number and per capita density of social services—such as senior centers, youth centers, food banks, job training programs, and day care centers—per United States census tract from 2003 through 2017.

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Data Sources

The North American Industry Classification System (NAICS) was used to identify social service organizations. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, including Puerto Rico and the US Virgin Islands. Other US island territories are excluded.

Methodology

This dataset highlights the concentration of social service organizations per census tract in the United States. Some of these organizations—senior centers in particular—function as common “third places” in the United States (Finlay et al., 2018), along with other gathering places such as gyms, churches, coffee shops, and retail stores. Third places are “physical locations outside

of the home (first place) or work/school (second place) that facilitate social interaction, community building, and social support” (Finlay et al., 2019). They are social infrastructure vital to civic life (Klinenberg, 2018). For other organizations that are not gathering spaces in the same way, such as food banks and emergency shelters, research has focused on the geographic distribution of these resources vis-à-vis their ability to serve their target communities (Bacon & Baker, 2017, Balcik & Beamon, 2008).

Our research team identified the following NAICS codes of interest:

- 624: all social services.
- 624110: child and youth services, including adoption centers, foster care placement services, and youth centers (excluding those that are for recreation only).
- 624120: services for the elderly and people with disabilities, such as senior centers, adult day care centers, and disability support groups.
- 624190: other individual and family services, such as suicide crisis centers, self-help centers for alcohol and drug addiction, and family social service agencies.
- 624210: community food services, i.e. food banks and food pantries.
- 624221: temporary shelters, such as for homeless individuals and victims of domestic violence. (Disaster relief shelters are excluded.)
- 624230: emergency relief services, such as disaster relief shelters and immigrant resettlement services.
- 624310: vocational rehabilitation services, such as job training and counseling, excluding schools.
- 624410: child day care services, including preschools, head start programs (separate from schools), in-home day cares, and before- and after-school care programs.

One other NAICS code, “other community housing services” (624229), was omitted because no results were found in the NETS database.

Note that NAICS codes are hierarchical, and three-digit codes are inclusive of all six-digit codes that start with the same pattern. For example, if a tract contains three social service organizations (624), one senior center (624120), one food pantry (624210), and one child day care (624410), the total number of establishments is three, not six.

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment’s census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](#). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](#). Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all social service organizations per tract in each year.

Usage Note

One type of social service organization, senior centers, are one of several common third places in communities. Data users interested in third places and community resources might find further useful data in these additional NaNDA datasets:

- [Arts, entertainment, and recreation organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020a), which includes gyms, libraries, and arcades, among others.
- [Eating and drinking places by census tract, United States, 2003-2017](#) (Esposito et al., 2020), which includes coffee shops, fast food restaurants, and bars.
- [Grocery stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020b).
- [Personal care services and laundromats by census tract, United States, 2003-2017](#) (Esposito et al., 2020b).
- [Post offices and banks by census tract, United States, 2003-2017](#) (Finlay et al., 2020c).
- [Religious, civic, and social organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020d).
- [Retail establishments by census tract, United States, 2003-2017](#) (Finlay et al., 2020e), which includes stores selling furniture, clothing, sporting goods, home and garden supplies, books, and used merchandise, among other items.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>.

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_624	int	1036462	157	6.172419	0	236	# all social services
count_sales_624	int	1036462	148	5.699564	0	223	# all social services w/ sales>0
count_emp_624	int	1036462	134	4.677498	0	203	# all social services w/ 2+ employees
popden_624	float	1015900	396025	88.89304	-1	3.50E+07	# all social services per 1000 people
popden_sales_624	float	1015869	387747	82.9482	-1	3.50E+07	# all social services per 1000 people w/ sales>0
popden_emp_624	float	1015819	364821	82.46727	-1	3.50E+07	# all social services per 1000 people w/ 2+ employees
aden_624	float	1017996	347296	7.937976	0	1328.507	# all social services per sq mile
aden_sales_624	float	1017996	340483	7.328126	0	1164.719	# all social services per sq mile w/ sales>0
aden_emp_624	float	1017996	313831	6.269577	0	1000.93	# all social services per sq mile w/ 2+ employees
count_624110	int	1036462	15	0.2178604	0	14	# child/youth services
count_sales_624110	int	1036462	15	0.2019428	0	14	# child/youth services w/ sales>0
count_emp_624110	int	1036462	14	0.1954186	0	13	# child/youth services w/ 2+ employees
popden_624110	float	1015154	76683	0.099007	-1	10064.75	# child/youth services per 1000 people
popden_sales_624110	float	1015152	72326	0.0920635	-1	10064.75	# child/youth services per 1000 people w/ sales>0
popden_emp_624110	float	1015148	70827	0.092615	-1	10064.75	# child/youth services per 1000 people w/ 2+ employees
aden_624110	float	1017996	28000	0.3322015	0	106.8382	# child/youth services per sq mile
aden_sales_624110	float	1017996	27497	0.3101434	0	91.79617	# child/youth services per sq mile w/ sales>0
aden_emp_624110	float	1017996	26497	0.3036791	0	106.8382	# child/youth services per sq mile w/ 2+ employees
count_624120	int	1036462	21	0.4145989	0	46	# elderly/disabled person services

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_sales_624120	int	1036462	19	0.3923106	0	46	# elderly/disabled person services w/ sales>0
count_emp_624120	int	1036462	20	0.3696527	0	46	# elderly/disabled person services w/ 2+ employees
popden_624120	float	1015184	111470	30.00751	-1	1.17E+07	# elderly/disabled person services per 1000 people
popden_sales_624120	float	1015175	107220	29.99883	-1	1.17E+07	# elderly/disabled person services per 1000 people w/ sales>0
popden_emp_624120	float	1015181	102800	29.99512	-1	1.17E+07	# elderly/disabled person services per 1000 people w/ 2+ employees
aden_624120	float	1017996	44590	0.4827337	0	117.9002	# elderly/disabled person services per sq mile
aden_sales_624120	float	1017996	44023	0.4565044	0	117.9002	# elderly/disabled person services per sq mile w/ sales>0
aden_emp_624120	float	1017996	42232	0.4438554	0	117.9002	# elderly/disabled person services per sq mile w/ 2+ employees
count_624190	int	1036462	121	2.35224	0	165	# other individual/family services
count_sales_624190	int	1036462	112	2.171537	0	158	# other individual/family services w/ sales>0
count_emp_624190	int	1036462	106	1.971105	0	144	# other individual/family services w/ 2+ employees
popden_624190	float	1015590	266271	55.78913	-1	2.34E+07	# other individual/family services per 1000 people
popden_sales_624190	float	1015557	256130	49.96425	-1	2.34E+07	# other individual/family services per 1000 people w/ sales>0
popden_emp_624190	float	1015556	243645	49.7206	-1	2.34E+07	# other individual/family services per 1000 people w/ 2+ employees
aden_624190	float	1017996	180903	3.392744	0	1128.321	# other individual/family services per sq mile
aden_sales_624190	float	1017996	176714	3.146236	0	982.7312	# other individual/family services per sq mile w/ sales>0
aden_emp_624190	float	1017996	171300	2.904869	0	837.1414	# other individual/family services per sq mile w/ 2+ employees
count_624210	int	1036462	4	0.0159379	0	3	# community food services
count_sales_624210	int	1036462	4	0.0152808	0	3	# community food services w/ sales>0
count_emp_624210	int	1036462	4	0.0130926	0	3	# community food services w/ 2+ employees
popden_624210	float	1015054	9281	0.0152496	-1	2937.33	# community food services per 1000 people
popden_sales_624210	float	1015054	9027	0.015073	-1	2937.33	# community food services per 1000 people w/ sales>0
popden_emp_624210	float	1015054	7694	0.0144784	-1	2937.33	# community food services per 1000 people w/ 2+ employees
aden_624210	float	1017996	1867	0.0183086	0	30.25475	# community food services per sq mile
aden_sales_624210	float	1017996	1863	0.0175097	0	30.25475	# community food services per sq mile w/ sales>0
aden_emp_624210	float	1017996	1668	0.0157341	0	30.25475	# community food services per sq mile w/ 2+ employees
count_624221	int	1036462	5	0.0274491	0	4	# temporary shelters
count_sales_624221	int	1036462	5	0.0257839	0	4	# temporary shelters w/ sales>0
count_emp_624221	int	1036462	5	0.024656	0	4	# temporary shelters w/ 2+ employees
popden_624221	float	1015050	14254	0.0113358	-1	294.0838	# temporary shelters per 1000 people

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popden_sales_624221	float	1015050	13528	0.0105887	-1	277.7577	# temporary shelters per 1000 people w/ sales>0
popden_emp_624221	float	1015047	13048	0.0105399	0	294.0838	# temporary shelters per 1000 people w/ 2+ employees
aden_624221	float	1017996	3421	0.0530812	0	62.65393	# temporary shelters per sq mile
aden_sales_624221	float	1017996	3403	0.0502384	0	62.65393	# temporary shelters per sq mile w/ sales>0
aden_emp_624221	float	1017996	3256	0.0488883	0	62.65393	# temporary shelters per sq mile w/ 2+ employees
count_624230	int	1036462	6	0.0342203	0	5	# emergency/relief services
count_sales_624230	int	1036462	6	0.0313943	0	5	# emergency/relief services w/ sales>0
count_emp_624230	int	1036462	6	0.0275398	0	5	# emergency/relief services w/ 2+ employees
popden_624230	float	1015084	17646	0.0219437	-1	2508.074	# emergency/relief services per 1000 people
popden_sales_624230	float	1015081	16480	0.021078	-1	2508.074	# emergency/relief services per 1000 people w/ sales>0
popden_emp_624230	float	1015072	14632	0.0177963	-1	2508.074	# emergency/relief services per 1000 people w/ 2+ employees
aden_624230	float	1017996	4904	0.0471663	0	73.68764	# emergency/relief services per sq mile
aden_sales_624230	float	1017996	4864	0.043506	0	73.68764	# emergency/relief services per sq mile w/ sales>0
aden_emp_624230	float	1017996	4459	0.0396448	0	73.68764	# emergency/relief services per sq mile w/ 2+ employees
count_624310	int	1036462	26	0.2459338	0	26	# vocational rehabilitation services
count_sales_624310	int	1036462	26	0.2306877	0	25	# vocational rehabilitation services w/ sales>0
count_emp_624310	int	1036462	21	0.1796998	0	21	# vocational rehabilitation services w/ 2+ employees
popden_624310	float	1015238	80854	30.08624	-1	1.17E+07	# vocational rehabilitation services per 1000 people
popden_sales_624310	float	1015232	77146	30.07817	-1	1.17E+07	# vocational rehabilitation services per 1000 people w/ sales>0
popden_emp_624310	float	1015217	66064	30.04433	-1	1.17E+07	# vocational rehabilitation services per 1000 people w/ 2+ employees
aden_624310	float	1017996	29841	0.316676	0	120.1844	# vocational rehabilitation services per sq mile
aden_sales_624310	float	1017996	29328	0.2967194	0	120.1844	# vocational rehabilitation services per sq mile w/ sales>0
aden_emp_624310	float	1017996	25950	0.2380504	0	120.1844	# vocational rehabilitation services per sq mile w/ 2+ employees
count_624410	int	1036462	61	2.891527	0	124	# child day care services
count_sales_624410	int	1036462	55	2.65635	0	109	# child day care services w/ sales>0
count_emp_624410	int	1036462	35	1.92279	0	36	# child day care services w/ 2+ employees
popden_624410	float	1015467	307502	2.743738	-1	507204.4	# child day care services per 1000 people
popden_sales_624410	float	1015451	298656	2.646461	-1	507204.4	# child day care services per 1000 people w/ sales>0
popden_emp_624410	float	1015371	265375	2.446959	-1	507204.4	# child day care services per 1000 people w/ 2+ employees
aden_624410	float	1017996	239620	3.33765	0	561.5981	# child day care services per sq mile

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
aden_sales_624410	float	1017996	234716	3.04714	0	561.5981	# child day care services per sq mile w/ sales>0
aden_emp_624410	float	1017996	204973	2.316504	0	459.4893	# child day care services per sq mile w/ 2+ employees

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